

Answer Key

Scientific Notation & Orders of Magnitude

1. The mass of the Earth is 5.97×10^{24} kg. What is the order of magnitude of this number? The mass of the Milky Way is about 6×10^{42} kg – by how many orders of magnitude is the mass of the Milky Way greater than the mass of the Earth? How many times do we need to multiply the mass of the Earth by 10 in order to reach the order of magnitude of the mass of the Milky Way?

Order of magnitude of Earth's mass: 10^{24} . Milky Way is $42 - 24 = 18$ orders of magnitude greater. Earth's mass must be multiplied by 10 a total of 18 times to reach the Milky Way's order of magnitude.

2. The age of the Universe is thought to be around 13787×10^6 years. How many significant figures does this measurement have? Write this number in proper scientific notation using *three* significant figures. Write the number again in scientific notation using *two* significant figures. (hint: when reducing the number of sig figs, you'll need to *round* the original measurement to the appropriate number of decimal places)

13787×10^6 years has 5 sig figs. The proper scientific notation with three sig figs is: 1.38×10^{10} years. With two sig figs: 1.4×10^{10} years

3. Your friend is writing a scientific paper about her cat. In her manuscript, she reports her cat's mass to be 4.536782 kg. Why might you be skeptical of this measurement? What additional information (e.g., about the equipment used to make this measurement) could your friend include in her paper that would make this number more believable? Write the cat's mass using three significant figures.

Most household devices do not have the precision of that many sig figs. You would need to know the precision of the device in order to believe the number. The cat's mass using three sig figs is: 4.54 kg

Arithmetic with Scientific Notation

4. Light travels at a speed of 1.8×10^7 kilometers per minute. The average distance from the Earth to the Sun is 1.5×10^8 kilometers. Find the time (in minutes) that it takes light to travel from the Sun to the Earth by *dividing* the Earth-Sun distance by the speed of light. Give your answer in scientific notation with the proper number of significant figures. If the Sun were to suddenly stop emitting light, how long would it take us (on Earth) to notice this?

$\frac{1.5 \times 10^8 \text{ km}}{1.8 \times 10^7 \text{ km/min}} = 8.3 \text{ min}$ (2 sig figs). It would take 8.3 min to notice the Sun has stopped emitting light since that is how long it takes light to travel from the Sun to the Earth.

5. The speed at which the Earth orbits around the Sun depends on the *sum* of the Earth's mass and the Sun's mass. The mass of the Earth is 5.972×10^{24} kg and the mass of the Sun is 1.989×10^{30} kg. Add these two masses together and express your answer in scientific

notation with the proper number of significant figures. Do you think the mass of the Earth play a significant role in determining the Earth's orbital speed around the Sun? Explain your answer.

$5.972 \times 10^{24} + 1.989 \times 10^{30} \approx 1.989 \times 10^{30}$ kg (4 sig figs). The Earth's mass is negligible compared to the Sun's, it does not meaningfully affect the orbital speed calculation.

Units and Unit Conversions

- Find someone you don't know in the class. Ask them their name, and when their birthday is. How old are they? Record your answer here. You should have the day, month, and year.
- Now, use unit conversion to figure out how many seconds old they are. Hint: There are 365 days in one year, 24 hours in a day, 60 minutes in an hour, and 60 seconds in one minute. You can use Google to see how many days it's been since their birthdate.

(days since birth) $\times 24 \times 60 \times 60 =$ age in seconds.

- The distance between us and the Andromeda Galaxy is 7.7×10^5 pc (pc = parsec). 1 parsec is equivalent to 3.09×10^{13} kilometers. What is the distance to the Andromeda Galaxy in units of *kilometers*? 1 parsec is also equivalent to 3.26 ly (ly = light years). What is the distance to the Andromeda Galaxy in units of *light years*? How many kilometers are in a light year?

$7.7 \times 10^5 \text{ pc} \times 3.09 \times 10^{13} \text{ km/pc} = 2.4 \times 10^{19} \text{ km}$ (2 sig figs).

In light years: $7.7 \times 10^5 \text{ pc} \times 3.26 \text{ ly/pc} = 2.5 \times 10^6 \text{ ly}$. Kilometers per light year: $\frac{2.4 \times 10^{19} \text{ km}}{2.5 \times 10^6 \text{ ly}} \approx 9.5 \times 10^{12} \text{ km/ly}$

- Based on your answer to the last question, what type of unit is a light year? Speed, distance, currency, mass, time, etc?

It's a distance.

- The average distance between the Earth and the Sun is defined as the *Astronomical Unit* (or AU), which is equivalent to 1.496×10^8 km. The average distance between the Sun and Jupiter is around 5.2 AU – what is the Jupiter-Sun distance in *kilometers*?

$5.2 \text{ AU} \times 1.496 \times 10^8 \text{ km/AU} = 7.8 \times 10^8 \text{ km}$

- The average speed at which the Earth orbits the Sun is around 29.8 km/s. What is the Earth's orbital speed in km/yr? What is the Earth's orbital speed in AU/yr? (1 yr = 3.15×10^7 s).

$29.8 \text{ km/s} \times 3.15 \times 10^7 \text{ s/yr} = 9.39 \times 10^8 \text{ km/yr}$.

In AU/yr: $\frac{9.39 \times 10^8 \text{ km/yr}}{1.496 \times 10^8 \text{ km/AU}} = 6.28 \text{ AU/yr}$

- Another common unit system used in astronomy is the centimeter / gram / second (CGS) system, where distances, masses, and time are respectively reported in these units. What is mass of the Earth in this unit system? What is the distance of the Earth to the Sun in this unit system?

Mass of Earth in CGS: $5.97 \times 10^{24} \text{ kg} \times 1000 \text{ g/kg} = 5.97 \times 10^{27} \text{ g}$. Distance from Earth to Sun in CGS: $1.496 \times 10^8 \text{ km} \times 10^5 \text{ cm/km} = 1.496 \times 10^{13} \text{ cm}$.

Parallax

13. The equation for D (the distance) in terms of x (the baseline, distance between points 1 and 2 in Figure 1) and θ (the full angle) using Figure 1.

$$D = \frac{x/2}{\tan(\theta/2)} \text{ (or, for small angles, } D \approx x/\theta \text{ with } \theta \text{ in radians)}$$

14. How does the angle change when you increase the baseline, x ?

Increasing the baseline x increases the parallax angle θ

15. How about when you increase the distance, D ?

Increasing the distance D decreases the parallax angle θ

16. **Calculate the length of your arm (in cm) using parallax. Does this make sense? Measure your arm using the meter stick. Were you close? Discuss.**

Students should find their calculated arm length (using $D = \frac{x/2}{\tan(\theta/2)}$, with x = eye separation and θ = measured angle) is reasonably close to their meter-stick measurement. Remind students to convert θ to radians if using the small-angle approximation.

17. Calculate the distance (in parsecs) to each of the four stars listed above.

Using $D [\text{pc}] = 1/p ["]$:

- Proxima Centauri: $1/0.768 = 1.30 \text{ pc}$
- Sirius: $1/0.379 = 2.64 \text{ pc}$
- Vega: $1/0.130 = 7.69 \text{ pc}$
- Betelgeuse: $1/0.0055 = 182 \text{ pc}$

18. Which star is closest to us? Which is farthest? Does this match your intuition about which parallax angles correspond to closer vs. farther objects?

Closest: Proxima Centauri. Farthest: Betelgeuse. Yes — this matches intuition, since a *larger* parallax angle corresponds to a *closer* star

19. 1 parsec is equivalent to about 3.26 light years. Convert your four distances into light years.

Converting to light years (distance in pc $\times$ 3.26):

- Proxima Centauri: $1.30 \times 3.26 = 4.25 \text{ ly}$
- Sirius: $2.64 \times 3.26 = 8.60 \text{ ly}$
- Vega: $7.69 \times 3.26 = 25.1 \text{ ly}$
- Betelgeuse: $182 \times 3.26 = 593 \text{ ly}$

20. Proxima Centauri is the closest known star to the Sun (other than the Sun itself). Does your calculated distance seem reasonable, given that light from the Sun takes about 8 minutes to reach Earth?

Yes, this is reasonable. Light from Proxima Centauri takes about 4.25 *years* to reach us, versus only 8 *minutes* from the Sun — consistent with Proxima being vastly farther away than the Earth-Sun distance, even though it's our nearest stellar neighbor.

21. Betelgeuse has a much smaller parallax angle than the other stars, making it harder to measure precisely. Based on what you found in Section 4.1 (how angle changes with distance), why might astronomers have more difficulty pinpointing the exact distance to farther stars like Betelgeuse?

A smaller parallax angle is harder to measure precisely (just as a very distant background object barely shifts when you switch eyes in Section 4.1). Since Betelgeuse's angle is so tiny, small measurement errors in θ translate into large uncertainties in the calculated distance D .